Clement (Kai Hei) Chu

Curriculum Vitae
`clement[dot]chu[at]mail[dot]utoronto.ca`
`https://clementc52.github.io/`

EDUCATION

University of Toronto
Mathematics and Its Applications Specialist in Probability and Statistics
Completion Year: May 2027
Relevant Coursework: Real Analysis, Complex Variables, Theory of Statistics, Theory of Statistical Inference, Statistical Computation, Time Series, Regression Analysis, Group Theory, Sampling Theory, Nonlinear Optimization Theory

RESEARCH INTERESTS

Probability theory, Generative modeling, Score-based and diffusion methods.

PUBLICATIONS

C. K. H. Chu. *Preservation of Baire Spaces under Homeomorphisms.* TRUNK Undergraduate Research Journal, 2025. [PDF]

EXPOSITORY PAPERS AND PROJECTS

Spectral Graph Theory (Directed Reading Project)
Department of Mathematics, University of Toronto Jan–May 2025
Supervisor: Narmada Varadarajan (PhD)

- Studied Perron-Frobenius theory, expander graphs, and pseudorandom graph constructions.
- Analyzed spectral properties of graph Laplacians and their applications to graph embeddings. [Slides]

Causal Inference (Directed Reading Project)
Department of Statistical Sciences, University of Toronto Jan–May 2026
Supervisor: Arian Hashmzadeh (PhD)

Expository Paper: Analysis of Robustness of Linear Estimators via ℓ^p and Dual Norm Lipschitz Bounds
Informally guided by PhD student *Jason (Sin Hang) Yeung*. [PDF]

Are Improper Priors Objective in Statistical Inference?
Course project for STA422/2162 under Prof. Michael Evans 2026

- Investigated the objectivity of improper priors, such as default and reference priors, with theoretical and philosophical perspectives. [PDF]

"A Note in Probability and Statistics" 2025-Present

- Developing expository notes on probability theory and statistical inference for early undergraduate audiences.

TEACHING EXPERIENCE

Private high school tutor in Mathematics 2023-Present

- Tutored high school students who are preparing for Cambridge IGCSE and IB public exams.

ADDITIONAL INFORMATION

Programming and Typesetting Languages: Python, R; LaTeX
Libraries: ggplot2, Quarto, NumPy
Awards: Dean's List Scholar (2024–2025)
Languages: English (Native), Cantonese (Native), Mandarin (Native), Japanese (Intermediate)